

DS 4002 Case Study

Can City Hall Move Housing Prices?

Policy Uncertainty and Housing Market Volatility in Charlottesville

The average apartment in Charlottesville now costs \$1,950 per month. Demand is high, supply is limited, and student-luxury developments are squeezing out public housing. Housing is one of the most pressing issues facing the city – and the people making decisions about it meet every month in a conference room on Fifth Street.

You’ve been hired as a data analyst for the City of Charlottesville. On your first day, your supervisor drops a question on your desk:

“When the Housing Advisory Committee sounds uncertain about policy – does the housing market react? Can we actually measure that?”

You have access to two data sources: the full text of every HAC meeting held in 2025, and monthly housing market reports from the Charlottesville Area Association of Realtors. Your job is to build a data pipeline that turns those meeting minutes into a quantitative uncertainty score – and then test whether that score predicts short-term swings in Charlottesville housing prices.

What You Will Do

Using Python, you will process text from government meeting minutes, construct an uncertainty score, merge it with real housing market data, and run a time-series regression. You will then write a short report interpreting your results – including being honest about what the data can and cannot tell us.

What You Will Deliver

- A written report (1-2 pages) summarizing your methodology, findings, and limitations
- Documented, reproducible Python code
- At least two visualizations of your data and results

Full details are in the rubric.